

Dispersion, not displacement

Loss of alveolar epithelial state coherence in lethal COVID-19

Pierce Taylor & Chimdi Walter Ndubuisi

Computational Systems Biology 8190

Dr. Toni Kazic

May 3, 2026

The question in one sentence

Do alveolar epithelial cells in lethal COVID-19 fail as a **coherent block** (pushed toward one damaged state), or do they **disperse** across many injury states?

- “Coherent block” $\Rightarrow$ expect **centroid displacement**.
- “Dispersion” $\Rightarrow$ expect **within-group variance inflation**.
- These predict *opposite* things — the failing test is informative.

Why this matters

- Melms 2021, Delorey 2021 already described alveolar injury in COVID-19 lungs.
- Our contribution is **methodological**:
 - Pre-specified geometric tests with opposing predictions
 - 10 systematic ablations to separate robust from fragile findings
 - Independent 618-donor cross-cohort replication
 - Donor-level modeling, portable state scores, mechanistic decomposition
- Honest reporting: which findings survive and which don't.

Biology recap (for non-biologists)

Two cell types line each air sac:

- **AT1:** thin, flat, gas exchange ($\sim 95\%$ of surface)
- **AT2:** secrete surfactant + serve as local stem cells

When AT1 die: AT2 $\rightarrow$ transitional (DATP)

$\rightarrow$ AT1 In lethal **COVID-19:**

- SARS-CoV-2 enters AT2 via ACE2
- Kills both surfactant factory AND stem cells
- Transitional cells accumulate (stuck mid-repair)

Key terms:

snRNA-seq: measure gene expression per nucleus

Pseudotime: computational ordering along a biological process

Dispersion: how spread out cells are in gene-expression space

DATP: damage-associated transient progenitor (stuck mid-transition)

Primary atlas (Melms 2021)

- SCP1219: 116,313 snRNA-seq nuclei
- 22,128 alveolar epithelial cells
- 7 healthy + 20 COVID-19 donors
- Full transcriptome ($\sim 20,000$ genes)

Replication (cellxgene census)

- 89,736 AT1/AT2 cells
- 618 donors, 35 datasets
- SCP1219 **excluded by ID**
- 84-gene panel only

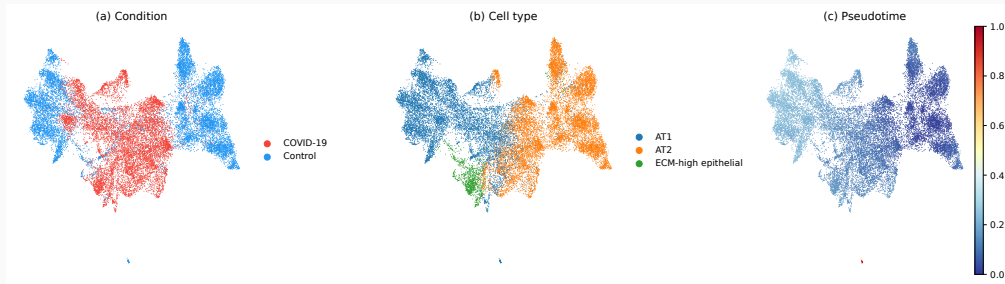
True sample size: 27 donors, not 22,128 cells.

Pipeline at a glance

1. QC: filter cells by gene count, mito %
2. Subset to alveolar epithelium (AT1 + AT2 + transitional)
3. Normalize (10,000 counts/cell) $\rightarrow \log(1 + x)$
4. 3,000 HVGs (Seurat v3) $\rightarrow$ Scale $\rightarrow$ 50-PC PCA
5. k -NN ($k=15$) $\rightarrow$ UMAP + 15-component diffusion map
6. Diffusion pseudotime rooted at healthy-AT2 centroid, normalized to $[0, 1]$
7. 8 curated gene programs scored via `sc.tl.score_genes`
8. 3 pre-specified statistical tests + 10 ablations

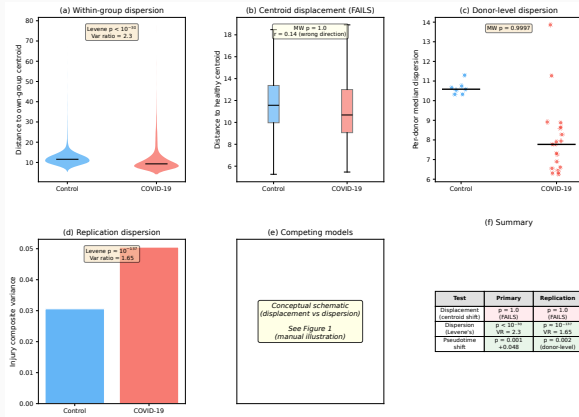
Software: Python 3.12.3, Scanpy 1.12.1, seed 42.

The manifold



UMAP colored by condition (left), cell type (center), pseudotime (right).

Result 1: Dispersion — the main finding



Within-group variance:

- COVID: 33.28
- Healthy: 14.58
- Ratio: 2.3×
- Levene $p < 10^{-15}$

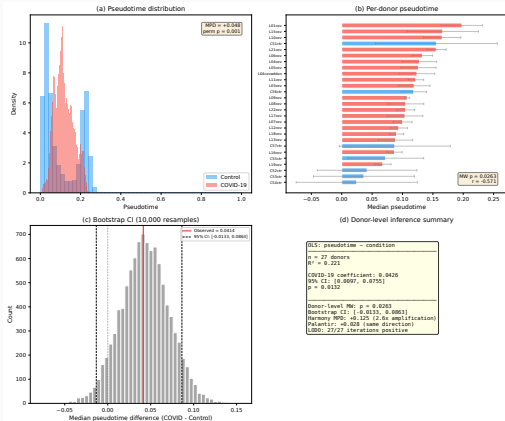
COVID cells occupy a 2.3-fold broader region of gene-expression space. **Replication**: ratio 1.65×, $p \approx 10^{-137}$ across 618 donors.

Result 2: Centroid displacement **fails**

- One-sided Mann–Whitney U (COVID > Healthy distance to healthy centroid):
 $p = 1.0$
- COVID median distance: 10.68 Healthy: 11.56
- COVID cells are actually *closer* to the healthy centroid!

This rules out the “one coherent damaged state” model. Why? Some COVID cells stay near homeostatic identity while others scatter to distant positions. A mean-displacement test averages over both and sees nothing.

Result 3: Pseudotime shift (supported)



- Median: COVID 0.111 vs Healthy 0.063
- $\Delta = +0.048$ ($\sim 5\%$ of range)
- Permutation $p = 0.001$
- LOO: 27/27 donors preserve direction
- Harmony amplifies $2.6\times$

Donor-level OLS:

$$\beta = +0.043, p = 0.013, R^2 = 0.22$$

Donor-level dispersion: a surprising twist

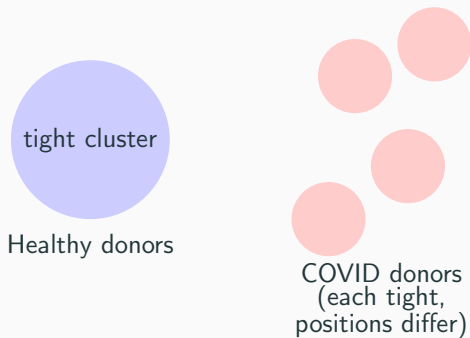
Cell level: COVID cells are $2.3\times$ more dispersed.

Donor level: COVID donors have **lower** within-donor dispersion (7.77 vs 10.58).

Resolution:

Individual COVID donors are *not* internally messier. The cell-level dispersion arises because **each COVID donor occupies a different region of injury space.**

Pooling across donors creates apparent dispersion.



Portable state scores

Marker-threshold DATP calling **fails cross-cohort**. We developed 6 continuous scores:

Score	What it measures	AUC (transitional)
AT2 identity	Homeostatic AT2 markers	0.323
AT1 identity	Homeostatic AT1 markers	0.375
Transitional	KRT8/CLDN4/SFN markers	0.441
Injury composite	Mean of 6 injury programs	0.630
Coherence loss	Distance from healthy centroid	0.972
Repair failure	Transitional – 0.5(AT2 + AT1)	0.730

All 6 scores: $p < 10^{-300}$ for condition separation in replication cohort.

Geometry-based scores transfer where marker thresholds fail.

Mechanistic decomposition: what drives the fan-out?

Which programs drive dispersion?

Linear regression: distance-to-healthy ~ 8
programs

$$R^2 = 0.115$$

Program	Coeff.
Senescence	+6.83
Oxidative stress	+3.83
NF- κ B	+1.95
AT2 identity	-1.13
AT1 identity	-1.02

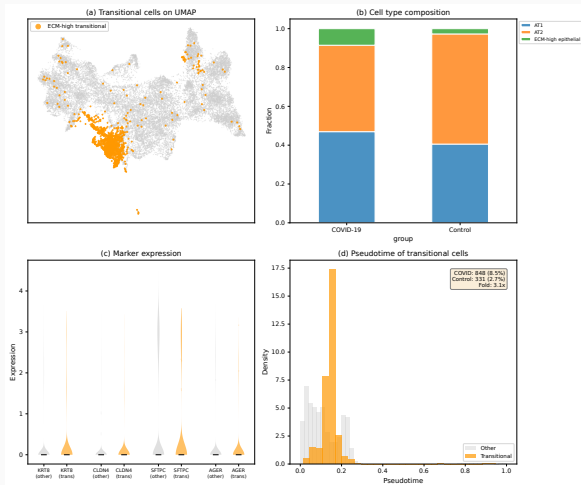
Interpretation:

The fan-out is driven by **divergent activation of stress/senescence programs** across cells.

Cells retaining homeostatic identity stay near the healthy centroid (negative coefficients).

Repair-stall: COVID cells initiate repair but are diverted toward apoptosis rather than completing AT1 differentiation.

Transitional cells: enriched but not portable



Primary atlas:

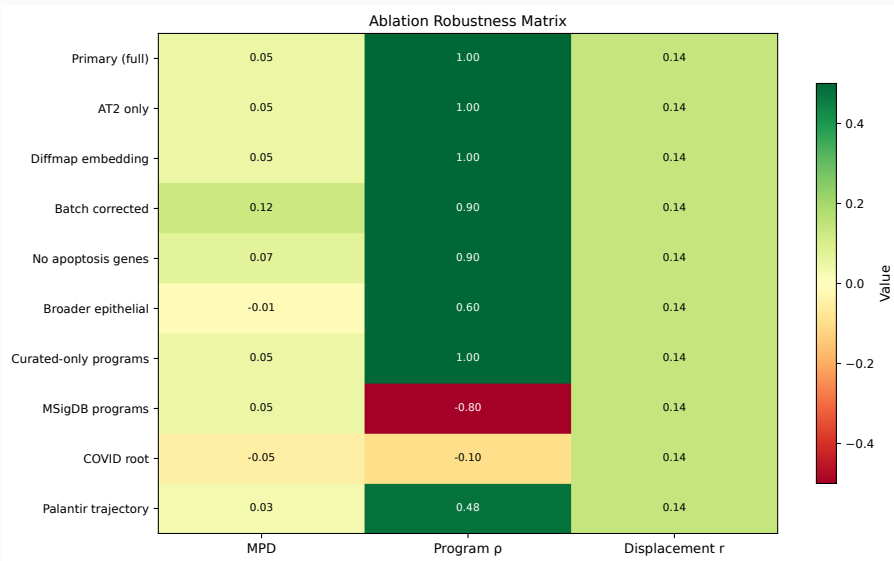
3.1 $\times$ enriched in COVID (8.5% vs 2.7%)

Replication:

KRT8⁺/CLDN4⁺ threshold gives *opposite* direction (0.48 $\times$)

Lesson: Marker-threshold cell-state definitions don't transfer across cohorts. Use manifold-based or classifier-based approaches.

Ten ablations — stress testing robustness



The Harmony bug — worth a slide

- Initially, corrected PCs were written with wrong orientation (PyTorch backend: $\text{cells} \times \text{PCs}$; old CPU backend: $\text{PCs} \times \text{cells}$)
- One-line fix: check shape and transpose if needed
- Post-fix: MPD $+0.048 \rightarrow +0.125$ (**$2.6\times$ amplification**)
- Moral: “batch correction didn’t help” can mean *the pipeline is broken*, not *the method doesn’t matter*

Independent replication: what transfers?

Finding	Primary	Replication (618 donors)	Geometric
Dispersion (Levene)	$p < 10^{-30}$, ratio $2.3\times$	$p \approx 10^{-137}$, ratio $1.65\times$	
Pseudotime direction	perm. $p = 0.001$	donor-level $p = 0.002$	
Centroid displacement	fails ($p = 1.0$)	fails ($p = 1.0$)	
KRT8 ⁺ /CLDN4 ⁺ DATP	$3.1\times$ enriched	$0.48\times$ — reversed	
State scores (6)	all significant	all $p < 10^{-300}$	

findings (dispersion, donor-level direction) are portable.

Annotation-dependent findings (marker thresholds, program ordering) are not.

Summary of evidence

	Primary	Ablations	Replication	Verdict
Dispersion	$p < 10^{-30}$	10/10 robust	$p \approx 10^{-137}$	Strong
Pseudotime shift	$p = 0.001$	27/27 LOO	donor $p = 0.002$	Robust
Donor-level OLS	$p = 0.013$	—	—	Supported
Centroid displacement	$p = 1.0$	10/10 fails	$p = 1.0$	Ruled out
DATP threshold	$3.1\times$	—	reversed	Not portable
Fine program order	varies	inverts	—	Fragile

The conclusion in one paragraph

Lethal COVID-19 does not push alveolar epithelial cells coherently away from the healthy state.

It **fans them outward** across heterogeneous injury states — driven by divergent senescence, oxidative stress, and inflammatory programs — while each donor's cells occupy a different region of injury space.

The strongest, most portable finding is **within-group dispersion**, which replicates across 35 datasets and 618 donors.

Key takeaways for computational systems biology

1. **Pre-register opposing hypotheses.** A failing test is informative.
2. **Dispersion $>$ centroid displacement** for heterogeneous disease states.
3. **Donor-level pseudobulk** beats cell-level tests ($n = 27$, not $n = 22,128$).
4. **Continuous state scores** beat marker thresholds for cross-cohort transfer.
5. **Ablations catch bugs.** The Harmony orientation fix is exhibit A.
6. **Report honestly** what fails, not just what succeeds.

Limitations

- Post-mortem, lethal cases only (no mild/moderate COVID)
- 27 donors is small — bootstrap CI marginally includes zero
- Pseudotime = ordering, not elapsed time
- 84-gene replication panel: too narrow for independent manifold
- Fine program ordering inverts under alternative gene lists
- DATP marker thresholds don't transfer cross-cohort
- Post-mortem effects may confound stress/apoptosis programs

Thank you.
Questions?